

CyberShot camera integrated into a phone—the CCD image sensor along with an 8X digital zoom snaps pictures of an excellent quality for a device which is “also a phone.” Add to that the tri-band phone capabilities, the HSCD and GPRS connectivity, Infrared and Bluetooth, an XHTML web browser, an FM radio, an MP3 player, and a video player. The math is simple—the S700i can perform ably as a camera, a phone and a PDA.

When a phone is designed to handle such multiple sources of data, storage becomes an essential element. Indeed, hard disk-based cell phones are trickling in. Samsung’s SPH-V5400 has a mega-pixel camera and a 1-inch, 1.5 GB hard disk drive. It was recently unveiled for the South Korean market. Also from Samsung is the SCH-S250, which offers a 5 MP CCD-based, digital camera-cum-camcorder, with a shutter speed of 1/1000th of a second, a macro mode of 10 cm and 92 MB of internal storage.

Where does that leave the PDA and the camera as discrete devices? For an average user, a convergent device will suffice and indeed replace discrete units. Will such convergent devices proliferate though? Phone carriers such as Airtel and Orange are known to subsidise their cost, so proliferation is solely in the hands of carriers, especially in price-sensitive markets such as India.

Thus, for the majority of us, a cell phone will eventually replace both the PDA and the camera.

TECHNOLOGY > DUAL-CORE PROCESSORS

Gigahertz Is Dead

“By mid-decade, the Pentium PC may need the power of a nuclear reactor. By the end of the decade, you might as well be feeling a rocket nozzle than touching a chip. And soon after 2010, PC chips could feel like the bubbly hot surface of the sun itself,” straight from the mouth of Patrick Gelsinger, Intel’s Chief Technology Officer.

Technology hasn’t been too kind to the processor, the heart of a computer, this year past. Faster chips have been crammed into shrinking spaces, resulting in monsters which greedily consume electricity and belch out enormous heat. The problems were so dramatic, especially for Intel; that a complete rethinking was in order—the 4 GHz Prescott was scrapped and the “Gigahertz is God” mantra unceremoniously dropped. After all, nobody wants the Sun inside their PC cabinet.

Thus, instead of making faster trucks to transport cargo, chipmakers will now offer a greater number of trucks which will move relatively sedately but in parallel, effectively performing the same work as their faster predecessors. This analogy translates to using multiple cores within a processor rather than using a single core running at 3+ GHz speeds.

A dual-core solution will be the herald of this thinking to our desktops, come next year. Such a solution effectively gives you two, nearly independent processors. Each CPU will execute different parts of a program. While this may not always double the speed of a single task, it will definitely improve overall performance.



The Treo 600 is amongst the most successful PDA-phone hybrids

EVENT > INDIAN IT COMPANIES CROSSING THE US \$1 BILLION MARK

The Billion Dollar Club

Last year saw the formation of the so-called Billion Dollar Club, then a collective of one, namely the privately-owned Tata Consultancy Services (TCS). This year witnessed the inculcation of two public companies—Infosys and Wipro Ltd. TCS’ revenues stand at US \$1.2 billion, Infosys’ at US \$1.08 billion and Wipro’s at US \$1.35 billion.

Why is this important? Firstly a billion dollar revenue is a good, healthy number, something a CEO of an IT company can proudly display on his visiting card, we imagine. More importantly, this three-member group, with billion dollar pockets each, will only seed and proliferate the Indian job market. TCS, for example has 24,000 employees working in 100 branches across 32 countries; while Infosys, as of mid-2004, employed 25,600 people.

While engineering and technology is taking away blue-collared jobs (the Indian automobile



TCS, Infosys and Wipro have all crossed US \$1 billion in terms of revenue

industry, for example recently underwent job cuts while simultaneously increasing production), the IT industry is hungry for more and is snapping up people faster than our colleges can churn them out. Recently, TCS, Infosys and Wipro jointly hired 14,000 new workers, almost as much as the entire workforce of Bajaj Auto and Maruti combined. IT sales are projected to rise between 30 and 35 per cent annually, which can only create more jobs.

There are other benefits—over 20 per cent of the employed workforce at Wipro, for example is comprised of women. IT is a good leveller.

Finally, if India has to put some money where its “IT superpower” mouth is, billion-dollar companies are an important milestone on the long road ahead.

PEOPLE > PRADIP BAIJAL, CHAIRMAN OF TRAI

Connecting People

Baijal Pradip, chairman of the Telecom Regulatory Authority of India (TRAI) had stepped into a mess. Wireless in Local Loop was the new kid on the block when he took charge and telecom



“After 2010, PC chips could feel like the bubbly hot surface of the sun itself”
Patrick Gelsinger, Intel’s CTO